

MATH20250 HW5 Observations

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Problem-Specific Observations

As always, also consult the debrief file on Canvas and the compiled example solutions from your fellow students.

LADW 4.2.3. Let

$$A = \begin{pmatrix} 4 & 3 \\ 1 & 2 \end{pmatrix}.$$

Find A^{2004} by diagonalizing A .

Necessarily, we first find the eigenvalues and eigenvectors of A . So, we have the following characteristic polynomial:

$$\det(A - \lambda I) = \begin{vmatrix} 4 - \lambda & 3 \\ 1 & 2 - \lambda \end{vmatrix} = (4 - \lambda)(2 - \lambda) - 3(1) = \lambda^2 - 6\lambda + 5.$$

Finding the roots, we have

$$\det(A - \lambda I) = 0 \Rightarrow \lambda^2 - 6\lambda + 5 = 0 \Rightarrow (\lambda - 1)(\lambda - 5) = 0 \Rightarrow \underline{\lambda = 1, 5}.$$

We now find the associated eigenvectors. Beginning with $\lambda = 1$, we have the following:

$$\begin{aligned} A - I &= \begin{pmatrix} 3 & 3 \\ 2 & 2 \end{pmatrix} \Rightarrow \left(\begin{array}{cc|c} 3 & 3 & 0 \\ 2 & 2 & 0 \end{array} \right) \xrightarrow[r_2 \rightarrow \frac{1}{2}r_2]{r_1 \rightarrow \frac{1}{3}r_1} \left(\begin{array}{cc|c} 1 & 1 & 0 \\ 1 & 1 & 0 \end{array} \right) \xrightarrow{r_2 \rightarrow r_2 - r_1} \left(\begin{array}{cc|c} 1 & 1 & 0 \\ 0 & 0 & 0 \end{array} \right) \\ &\Rightarrow \text{Eigenvector for } \lambda = 1 : \begin{pmatrix} -1 \\ 1 \end{pmatrix}. \end{aligned}$$

Now for $\lambda = 5$:

$$\begin{aligned} A - 5I &= \begin{pmatrix} -1 & 3 \\ 1 & -3 \end{pmatrix} \Rightarrow \left(\begin{array}{cc|c} -1 & 3 & 0 \\ 1 & -3 & 0 \end{array} \right) \xrightarrow{r_1 \leftrightarrow r_2} \left(\begin{array}{cc|c} 1 & -3 & 0 \\ -1 & 3 & 0 \end{array} \right) \xrightarrow{r_2 \rightarrow r_2 + r_1} \left(\begin{array}{cc|c} 1 & -3 & 0 \\ 0 & 0 & 0 \end{array} \right) \\ &\Rightarrow \text{Eigenvector for } \lambda = 5 : \begin{pmatrix} 3 \\ 1 \end{pmatrix}. \end{aligned}$$

So, we observe the following diagonalization of A :¹

$$A = \begin{pmatrix} -1 & 3 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 5 \end{pmatrix} \begin{pmatrix} -1 & 3 \\ 1 & 1 \end{pmatrix}^{-1} = \begin{pmatrix} -1 & 3 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 5 \end{pmatrix} \begin{pmatrix} -\frac{1}{4} & \frac{3}{4} \\ \frac{1}{4} & \frac{1}{4} \end{pmatrix}.$$

Let $n \in \mathbb{R}$. Recall that, if $A = PDP^{-1}$, we have that

$$A^n = \underbrace{(PDP^{-1}) \cdots (PDP^{-1})}_n = PD^nP^{-1}.$$

Recall that, for a diagonal matrix D , D^n is the diagonal matrix with all entries of D raised to the n th power. This means that

$$\begin{aligned} A^{2004} &= \begin{pmatrix} -1 & 3 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 5 \end{pmatrix}^{2004} \begin{pmatrix} -\frac{1}{4} & \frac{3}{4} \\ \frac{1}{4} & \frac{1}{4} \end{pmatrix} = \begin{pmatrix} -1 & 3 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 5^{2004} \end{pmatrix} \begin{pmatrix} -\frac{1}{4} & \frac{3}{4} \\ \frac{1}{4} & \frac{1}{4} \end{pmatrix} \\ &= \begin{pmatrix} -1 & 3 \cdot 5^{2004} \\ 1 & 5^{2004} \end{pmatrix} \begin{pmatrix} -\frac{1}{4} & \frac{3}{4} \\ \frac{1}{4} & \frac{1}{4} \end{pmatrix} \Rightarrow A = \begin{pmatrix} \frac{1+3 \cdot 5^{2004}}{4} & \frac{3+3 \cdot 5^{2004}}{4} \\ \frac{-1+5^{2004}}{4} & \frac{3+5^{2004}}{4} \end{pmatrix} = \frac{1}{4} \begin{pmatrix} 1+3 \cdot 5^{2004} & 3+3 \cdot 5^{2004} \\ -1+5^{2004} & 3+5^{2004} \end{pmatrix}. \end{aligned}$$

LADW 4.2.8. Find all square roots of the matrix

$$\begin{pmatrix} 5 & 2 \\ -3 & 0 \end{pmatrix}$$

i.e. find all matrices B such that $B^2 = A$. Hint: Finding a square root of a diagonal matrix is easy. You can leave your answer as a product.

Per the hint, we want to think about how diagonal matrices can help us here. Remember that

$$B^2 = A \Leftrightarrow B = A^{\frac{1}{2}}.$$

As such, we can diagonalize A – assuming it is diagonalizable – and find its square roots from there. To diagonalize, we find the eigenvalues of A :

$$\det(A - \lambda I) = \begin{vmatrix} 5 - \lambda & 2 \\ -3 & -\lambda \end{vmatrix} = (5 - \lambda)(-\lambda) - 2(-3) = \lambda - 5\lambda + 6.$$

Finding the roots, we have

$$\det(A - \lambda I) = 0 \Rightarrow \lambda^2 - 5\lambda + 6 = 0 \Rightarrow (\lambda - 2)(\lambda - 3) = 0 \Rightarrow \underline{\lambda = 2, 3}.$$

We now find the associated eigenvectors. Beginning with $\lambda = 2$, we have the following:

$$A - 2\lambda = \begin{pmatrix} 3 & 2 \\ -3 & -2 \end{pmatrix} \Rightarrow \left(\begin{array}{cc|c} 3 & 2 & 0 \\ -3 & -2 & 0 \end{array} \right) \xrightarrow{r_2 \rightarrow r_2 + r_1} \left(\begin{array}{cc|c} 3 & 2 & 0 \\ 0 & 0 & 0 \end{array} \right)$$

¹You can also use the diagonalization $A = \begin{pmatrix} 3 & -1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 5 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 3 & -1 \\ 1 & 1 \end{pmatrix}^{-1}$. Remember, if $A = PDP^{-1}$, where D is the diagonal matrix of eigenvalues of A , make sure the first column of P is an eigenvector of the eigenvalue in the first row and column of D , and so on.

$$\Rightarrow \text{Eigenvector for } \lambda = 2 : \begin{pmatrix} 2 \\ -3 \end{pmatrix}.$$

Now for $\lambda = 3$:

$$A - 3I = \begin{pmatrix} 2 & 2 \\ -3 & -3 \end{pmatrix} \Rightarrow \left(\begin{array}{cc|c} 2 & 2 & 0 \\ -3 & -3 & 0 \end{array} \right) \xrightarrow[r_2 \rightarrow \frac{1}{3}r_2]{r_1 \rightarrow \frac{1}{2}r_1} \left(\begin{array}{cc|c} 1 & 1 & 0 \\ -1 & -1 & 0 \end{array} \right) \xrightarrow{r_2 \rightarrow r_2 + r_1} \left(\begin{array}{cc|c} 1 & 1 & 0 \\ 0 & 0 & 0 \end{array} \right)$$

$$\Rightarrow \text{Eigenvector for } \lambda = 3 : \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

So, we observe the following diagonalization of A :

$$A = \begin{pmatrix} 2 & 1 \\ -3 & -1 \end{pmatrix} \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix} \begin{pmatrix} 2 & 1 \\ -3 & -1 \end{pmatrix}^{-1} = \begin{pmatrix} 2 & 1 \\ -3 & -1 \end{pmatrix} \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix} \begin{pmatrix} -1 & -1 \\ 3 & 2 \end{pmatrix}.$$

To calculate $A^{\frac{1}{2}}$, we note that

$$\begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}^{\frac{1}{2}} = \begin{pmatrix} \pm\sqrt{2} & 0 \\ 0 & \pm\sqrt{3} \end{pmatrix}.$$

Important: This means there are **four** (4) matrices C such that $C^2 = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}$. At any rate, we have

$$A^{\frac{1}{2}} = \begin{pmatrix} 2 & 1 \\ -3 & -1 \end{pmatrix} \begin{pmatrix} \pm\sqrt{2} & 0 \\ 0 & \pm\sqrt{3} \end{pmatrix} \begin{pmatrix} -1 & -1 \\ 3 & 2 \end{pmatrix}.$$

Expanding the products, we have

$$\begin{aligned} \begin{pmatrix} 2 & 1 \\ -3 & -1 \end{pmatrix} \begin{pmatrix} \sqrt{2} & 0 \\ 0 & \sqrt{3} \end{pmatrix} \begin{pmatrix} -1 & -1 \\ 3 & 2 \end{pmatrix} &= \begin{pmatrix} -2\sqrt{2} + 3\sqrt{3} & -2\sqrt{2} + 2\sqrt{3} \\ 3\sqrt{2} - 3\sqrt{3} & 3\sqrt{2} - 2\sqrt{3} \end{pmatrix} \\ \begin{pmatrix} 2 & 1 \\ -3 & -1 \end{pmatrix} \begin{pmatrix} -\sqrt{2} & 0 \\ 0 & \sqrt{3} \end{pmatrix} \begin{pmatrix} -1 & -1 \\ 3 & 2 \end{pmatrix} &= \begin{pmatrix} 2\sqrt{2} + 3\sqrt{3} & 2\sqrt{2} + 2\sqrt{3} \\ -3\sqrt{2} - 3\sqrt{3} & -3\sqrt{2} - 2\sqrt{3} \end{pmatrix} \\ \begin{pmatrix} 2 & 1 \\ -3 & -1 \end{pmatrix} \begin{pmatrix} \sqrt{2} & 0 \\ 0 & -\sqrt{3} \end{pmatrix} \begin{pmatrix} -1 & -1 \\ 3 & 2 \end{pmatrix} &= \begin{pmatrix} -2\sqrt{2} - 3\sqrt{3} & -2\sqrt{2} - 2\sqrt{3} \\ 3\sqrt{2} + 3\sqrt{3} & 3\sqrt{2} + 2\sqrt{3} \end{pmatrix} \\ \begin{pmatrix} 2 & 1 \\ -3 & -1 \end{pmatrix} \begin{pmatrix} -\sqrt{2} & 0 \\ 0 & -\sqrt{3} \end{pmatrix} \begin{pmatrix} -1 & -1 \\ 3 & 2 \end{pmatrix} &= \begin{pmatrix} 2\sqrt{2} - 3\sqrt{3} & 2\sqrt{2} - 2\sqrt{3} \\ -3\sqrt{2} + 3\sqrt{3} & -3\sqrt{2} + 2\sqrt{3} \end{pmatrix} \end{aligned}$$

as square roots of A .

LADW 4.2.10. Let A be a 5×5 matrix with 3 eigenvalues (not counting multiplicities). Suppose we know that one eigenspace is three-dimensional. Can you say if A is diagonalizable?

Recall the **criterion of diagonalizability**, given in Theorem 2.8 of LADW:

Let an operator $A : V \rightarrow V$ has exactly $n = \dim(V)$ eigenvalues (counting multiplicities). Then A is diagonalizable if and only if for each eigenvalue λ **the dimension of the eigenspace $\ker(A - \lambda I)$ (i.e. the geometric multiplicity of λ) coincides with the algebraic multiplicity of λ .**

By the fundamental theorem of algebra, we know that A must have 5 eigenvalues, counting multiplicities. From the problem, we know that $\sigma(A) = \{\lambda_1, \lambda_2, \lambda_3\}$. So,

$$\det(A - \lambda I) = (\lambda_1 - \lambda)^x (\lambda_2 - \lambda)^y (\lambda_3 - \lambda)^z,$$

where algebraic multiplicities x , y , and z are positive integers that sum to 5.

Without loss of generality, suppose λ_1 is the eigenvalue whose corresponding eigenspace is three-dimensional. That is,

$$\dim(\ker(A - \lambda_1 I)) = 3.$$

Recall Proposition 4.1.1 on pg. 103 of LADW:

Geometric multiplicity of an eigenvalue **cannot exceed** its *algebraic* multiplicity.

As such, $x \geq 3$. In order for λ_2 and λ_3 to be eigenvalues of A , we must have that $y, z \geq 1$, meaning

$$x + y + z \geq 3 + 1 + 1 = 5 \Rightarrow x + y + z \geq 5 \Rightarrow x + y + z = 5 \Rightarrow x = 3, y = 1, z = 1.$$

Necessarily, λ_1 must have algebraic multiplicity exactly 3, while λ_2 and λ_3 must each have algebraic multiplicity 1,² meaning

$$\dim(\ker(A - \lambda_2 I)) \leq 1 \quad \text{and} \quad \dim(\ker(A - \lambda_3 I)) \leq 1.$$

Recall that the dimension of a vector space must be a nonnegative integer; suppose one of the above eigenspaces had dimension 0. That eigenspace would be precisely the zero vector – however, in order for a constant λ to be an eigenvector of A , the vector $\mathbf{v}$ for which $A\mathbf{v} = \lambda\mathbf{v}$ is satisfied must be nonzero.

So, it must be that

$$\dim(\ker(A - \lambda_2 I)) = 1 \quad \text{and} \quad \dim(\ker(A - \lambda_3 I)) = 1.$$

Therefore, since each eigenvalue of A has algebraic multiplicity equal to its geometric multiplicity, **A is diagonalizable.**

²Suppose not. If algebraic multiplicity of λ_1 $x > 3$, at least one of y and z , the algebraic multiplicities of λ_2 and λ_3 , would have to be 0; in that case, that “eigenvalue” would not, in fact, be an eigenvalue.

General Observations

IMPORTANT: Any problems with this boxed “x” next to them – **X** – should NOT be chosen for your HW submission. If you do this on one HW assignment, you will be warned but given credit. On subsequent assignments, you will not receive credit for problems that have been so marked.

- **Observations on Proofs**

- (a) **Directionality:** Suppose you have to prove “ A if and only if B .” ($A \Leftrightarrow B$) This is a **twofold** process:
 - i. **Prove “if A , then B .”** This may be done by supposing A is true and proving B directly, supposing A is true but B is false and proving by contradiction, or by contrapositive – “if A , then B ” is logically equivalent to “if not B , then not A .”
 - ii. **Prove “if B , then A .”** This may be done by supposing B is true and proving A directly, supposing B is true but A is false and proving by contradiction, or by contrapositive – “if B , then A ” is logically equivalent to “if not A , then not B .”
- (b) **Defining variables and other quantities:** Whenever you introduce a new quantity, be it a scalar, vector, etc., **state the space in which it lives.**

Ex. $\mathbf{v} \in V$ (where V is a vector space), $\alpha \in \mathbb{C}$.

The $\in$ symbol, meaning “is an element of,” is generated by `\in`.

- (c) **Existence:** If you are tasked with proving the existence of a quantity, **do not presuppose its existence in your notation.**
 - i. Example: LADW 1.6.7 – The goal is to prove that the matrix B is invertible given that the matrices A and AB are invertible. It may be tempting at some point in the proof to use the following fact from Theorem 6.3 (pg. 26):

$$(AB)^{-1} = B^{-1}A^{-1}.$$

However, there is a key condition attached to that fact:

If linear transformations A and B are **invertible**...

As such, this fact should not be employed unless you have already shown that A and B are invertible.

- (d) **Justifications:** There are two main things to remember when justifying the steps in a proof:
 - i. **Prove all facts on which you rely.** If you need to use a result that you did not already prove in class, **prove it explicitly.** If you need to use a result that comes from another homework problem, **prove it explicitly.** Do not take other homework problems for granted – see LADW 4.1.9.
 - ii. If you are unsure whether to justify something explicitly, **do it.** Better safe than sorry.

- Miscellaneous

- (a) **Check your calculations.** Especially beware of dropping negative signs and accidentally performing the wrong arithmetic operation on two numbers (e.g., adding instead of multiplying).
 - i. When diagonalizing a matrix A , **supply the full $A = PDP^{-1}$ equation.** Additionally, make sure you order the eigenvalues in D to correspond with the proper eigenvectors in P .
 - ii. If you are finding a solution to a linear system of equations, **substitute your answer to check.** Sometimes, after converting to matrix form, it can be difficult to remember that there was an underlying group of equations. Use them to your advantage.
 - iii. If you are decomposing a matrix (e.g., writing it as a product of elementary matrices), **multiply the resulting matrices** to make sure you get the result you expect.
- (b) **Read the problems carefully.** Make sure you answer **every part of the problem.**
 - i. This also means you should not make any additional assumptions than what the problem allows.
 - ii. Additionally, for true/false questions, if a statement is true under a certain set of circumstances but *not under the set of circumstances outlined* in the statement, **the statement is false.** If no more specifics are given, err on the side of generality.